MILITARY SPECIFICATION

PACKING ASSEMBLIES, HYDRAULIC CONICAL AND V TYPES, FIRE RESISTANT TYPE HYDRAULIC FLUIDS

1. SCOPE

1.1 Scope. - This specification covers packing assemblies for safety type hydraulic fluid service such as elevators, hoists and steering gears.

1.2 Classification. - Hydraulic packing assemblies, shall be of the following types, as specified (see 6.1):

Type I - Hydraulic conical, symbol 1230.
Type II - Hydraulic V, symbol 1240.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Fiber.
PPP-B-676 - Boxes, Set-Up, Paperboard.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant.

MILITARY
MIL-P-116 - Preservation, Methods of.
MIL-B-4229 - Boxes; Paperboard, Metal-Stayed.
MIL-B-10377 - Box: Wood-Cleated, Veneer, Paper Overlaid.
MIL-L-10547 - Liners, Case, Waterproof.
MIL-H-19457 - Hydraulic Fluid, Fire Resistant

NAVY DEPARTMENT
General Specifications for Inspection of Material.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes, and Appendix - Sampling for Expensive Testing by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

FSC 5330

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2.2 Other publications. - The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St., New York 16, N.Y.)

3. REQUIREMENTS

3.1 Qualification. - Packing material furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein (see 6.2).

3.2 Composition and construction. - The composition and construction shall conform to that of the sample given qualification approval (see 4.2).

3.3 Size. - Packing shall be furnished in the sizes specified (see 6.1).

3.4 Graphite lubrication. - Graphite lubrication, when used, shall not exceed 1 percent, by weight, of the finished packing.

3.5 Compound. -

3.5.1 Woven cotton fabric. - The friction compound for woven cotton fabric packing shall be not more than 60 percent, by weight, of the finished packing. The compound shall contain not less than 20 percent, by weight, of synthetic rubber.

3.5.2 Leather. - The leather shall contain not less than 35 percent, by weight, of synthetic rubber impregnating compound.

3.6 Simulated performance. - The packing during initial conditions and during progress of the simulated performance test specified in 4.5.2 shall respond readily to gland adjustment for the control of leakage from the stuffing box. The packing shall be in satisfactory condition upon completion of test.

3.7 Type I - hydraulic conical, symbol 1230. -

3.7.1 Construction. -

3.7.1.1 Woven cotton fabric. - Packing shall consist of conical or beveled shape rings which permit each ring to nest within the adjacent ring, and one top and bottom adapter ring. The ring and adapters shall be composed of layers of woven cotton fabric, and treated with a synthetic rubber friction compound.

3.7.1.1.1 Fabric. - The cotton fabric shall be not less than 40 percent, by weight, of the finished packing.

3.7.1.2 Leather. - Leather packing shall consist of conical or beveled shape rings which permit each ring to nest within the adjacent ring, and one top and bottom adapter ring. The leather ring and adapters shall be impregnated with a synthetic rubber compound.

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3.8 Type II - hydraulic V, symbol 1240. -

3.8.1 Construction. -

3.8.1.1 Woven cotton fabric. - Packing shall consist of either V, U, or C cross-sectional shape rings which permit each ring to nest within the adjacent ring, and one top and bottom adapter ring. The rings and adapters shall be composed of layers of either cotton, asbestos or wire-inserted asbestos fabric, and treated with a synthetic rubber friction compound. Where wire-inserted asbestos cloth is used in the construction, it shall be so placed that it will bear against the wall of the stuffing box and not against the moving member.

3.8.1.1.1 Fabric. - The cotton fabric shall be not less than 40 percent, by weight of the finished packing.

3.9 Workmanship. - Workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.2 Qualification tests at a laboratory. - Qualification tests shall be conducted at a laboratory designated by the Bureau of Ships. These tests shall consist of the tests specified in 4.5.

4.3 Sampling. -

4.3.1 Lot. - For purposes of sampling, a lot shall consist of all packing of one type and size produced in one plant under essentially the same conditions and offered for delivery at one time.

4.3.2 Sampling for visual and dimensional examination. - A random sample of sets shall be selected from each lot of material offered for Government inspection in accordance with Standard MIL-STD-105 at inspection level III, and shall be subjected to the inspection specified in 4.4.1. The Acceptable Quality Level shall be 2.5 percent defective.

4.3.3 Sampling for lot acceptance tests. - A random sample of sets shall be selected from each lot offered for Government tests in accordance with the appendix to Standard MIL-STD-105 at inspection level L-4. In addition, a 12-inch square sample of untreated cotton or asbestos fabric shall be selected from each lot used in the manufacture of the finished packing. The sample sets and 12-inch square shall be forwarded to the Government laboratory for testing specified in 4.4.2.

4.4 Examination and tests. -

4.4.1 Visual and dimensional examination. - Each of the sample sets selected in accordance with 4.3.2 shall be visually and dimensionally examined to verify compliance with the requirements of this specification. Any set in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective sets in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4.2 Lot acceptance tests at Government laboratory. - The samples selected in accordance with 4.3.3 shall be subjected to those tests deemed necessary by the Government laboratory to determine that the samples conform with that given quallfication. If any sample set, or the 12-inch square, fails any test the lot represented by the sample shall be rejected.

4.4.2.1 Place of tests. - Unless otherwise specified in the contract or order, tests shall be conducted at the U.S. Naval Engineering Experiment Station, Annapolis, Md.

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4.4.3 Rejected lots. - Rejected lots may be offered again for Government inspection provided the contractor has repaired or removed all nonconforming material. The Government inspector shall again select and examine samples from such resubmitted lots to verify compliance with the specification requirements.

4.5 Test procedures. -

4.5.1 Chemical analysis. - Chemical analysis shall be made using standardized procedures.

4.5.2 Simulated performance (see 3.6). - Performance tests shall be made under simulated operating conditions in a machine designed for the purpose, as follows:

4.5.2.1 The apparatus consists of a 6-1/2-inch diameter hydraulic ram fitted with a cylinder at one end and a 5/8-inch diameter valve stem at the other end. The ram has a stroke of 6 inches and makes 12 complete forward and return cycles per minute. The valve stem has a stroke of 2 inches and makes 12 complete forward and return cycles per minute. The packing shall be initially tested for tightness under hydrostatic pressure of 2,500 pounds per square inch gage, the rams and valves at rest. The packings are then subjected to safety type hydraulic fluid conforming to Specification MIL-H-19457 at a temperature of 150° F., and a pressure of approximately 1,500 pounds per square inch gage. The change from maximum to minimum pressures occurs every stroke of the ram. The apparatus is operated for a total of 1,000 hours.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (see 6.1). -

5.1.1 Level A. - The packing shall be individually preserved and packaged in accordance with method 1A-8 of Specification MIL-P-116. Contact preservative is not required.

5.1.1.1 When more than 10 identical items are being shipped at the same time to one destination, the items shall be intermediate packaged in quantities of not more than 10 in an intermediate package. Intermediate containers shall conform to Specification PPP-B-566, PPP-B-676, MIL-B-4229, or PPP-B-636, at the option of the contractor. Box closures shall conform to the applicable box specification and the appendix thereto. The gross weight of paperboard boxes shall not exceed 10 pounds and of fiberboard boxes shall not exceed 20 pounds. When 10, or less, identical items are being shipped at the same time to one destination, intermediate packaging will not be required and the item shall be packaged as specified herein and packed directly into shipping containers as specified for level A, B, or C, (see 5.2) as applicable.

5.1.2 Level C. - The packing shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing (see 6.1). -

5.2.1 Level A. - The packing, preserved and packaged as specified (see 6.1), shall be packed in overseas type wood cleated fiberboard, nailed wood, wirebound wood, corrugated or solid fiberboard, wood cleated paper overlaid, or wood cleated plywood boxes conforming to Specification PPP-B-591, PPP-B-621, PPP-B-585, PPP-B-636 (class 3), MIL-B-10377 or PPP-B-601, respectively, at the option of the contractor. Shipping containers shall have caseliners conforming to Specification MIL-L-10547 and shall be closed and sealed in accordance with the appendix thereto. Caseliners for boxes conforming to Specification PPP-B-636 may be omitted provided all joints and corners of the boxes are sealed with a minimum of 1-1/2 inches of pressure sensitive tape conforming to Specification PPP-T-76. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

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5.2.2 Level B. - The packing, preserved and packaged as specified (see 6.1), shall be packed in domestic type wood cleated fiberboard, nailed wood, wirebound wood, cleated plywood or wood cleated paper overlaid boxes or class 2 fiber boxes conforming to Specification PPP-B-591, PPP-B-621, PPP-B-585, PPP-B-601, MIL-B-10377, or PPP-B-636, respectively, at the option of the contractor. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.3 Level C. - The packing, preserved and packaged as specified (see 6.1), shall be packed in containers which will insure acceptance and safe delivery at destination. Shipping containers shall comply to the Uniform Freight Classification Rules or other regulations as applicable to the mode of transportation.

5.3 Marking. - In addition to any special marking required by the contract, order or herein, interior and exterior shipping container shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size required (see 1.2 and 3.3).
(c) Whether packing shall be packaged in accordance with level A or C; packed for shipment in accordance with level A, B, or C (see 5.1 and 5.2).

6.2 Qualification. - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL-21099, whether or not such products have actually been so listed by that date.

6.2.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D. C.

6.3 The packaging, packing, and marking specified herein apply only to direct shipment to the Government and are not intended to apply to contracts or orders between the manufacturer and the prime contractor.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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